

Drifting with Intent: Generative Interest Trajectories for Multimodal Web Recommendation

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Abstract

User interest is not static but rather a continuously evolving process shaped by diverse interactions across multimodal web platforms. Traditional recommendation systems often model user preferences as discrete snapshots or oversimplified trajectories, failing to capture the nuanced dynamics of interest evolution. In this work, we propose *Drifting with Intent*, a novel framework that formalizes user interest as a continuous-time stochastic process governed by spatio-temporal coupled stochastic differential equations (SDEs). Our approach fundamentally departs from diffusion-based methods by modeling interest evolution as a directed drift toward meaningful content rather than random diffusion. The core innovation lies in a multi-granularity hypergraph encoder that captures cross-scale user-item interactions, coupled with a unified SDE solver that generates personalized interest trajectories across temporal and relational dimensions. Unlike prior work employing redundant twin networks, we introduce a generative-discriminative co-optimization framework that efficiently balances content generation and recommendation precision within a single parameter space. Extensive experiments across eight diverse datasets demonstrate that our framework not only outperforms state-of-the-art methods in recommendation quality but also provides interpretable interest trajectories that reveal how user preferences evolve across different modalities and time scales. Our work bridges the gap between generative modeling and practical recommendation systems by treating user interest as a purposeful drift rather than a random walk.

CCS Concepts

• **Information systems** → **Recommender systems**; *Social networks*; • **Computing methodologies** → *Neural networks*.

Keywords

Generative Recommendation, Stochastic Differential Equations, Multimodal Representation, User Interest Modeling, Sequential Recommendation, Web Mining

ACM Reference Format:

Weilin Zhou, Hao Zhang, and Guangxin Wu. 2026. Drifting with Intent: Generative Interest Trajectories for Multimodal Web Recommendation. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3774904.3792375>

1 Introduction

User interests on the web are inherently dynamic processes shaped by continuous interactions across multiple modalities—videos, social connections, e-commerce behaviors, and more. While traditional recommendation systems often treat user preferences as static entities or discrete state transitions, this simplification fails to capture the nuanced evolution of interests over time. The emerging paradigm of diffusion models has attempted to address this limitation by modeling interest evolution as a noise-addition and denoising process. However, these approaches fundamentally misunderstand the nature of user interest evolution: interests don't randomly diffuse but rather drift purposefully toward meaningful content with clear intent.

To empirically validate this continuous nature, we analyze the behavioral patterns in real-world datasets. As illustrated in Figure 1, the distance between consecutive interactions in the embedding space is predominantly small, suggesting that user preferences evolve along a coherent trajectory rather than making disjointed jumps.

In this work, we propose ST-SDERec (Spatio-Temporal Stochastic Differential Equation Recommender), a novel framework that formalizes user interest evolution as a continuous-time stochastic process governed by purposeful drift rather than random diffusion. Unlike RecDiff's [5] diffusion-based approach that requires twin networks for generation and discrimination, our framework unifies these objectives within a single parameter space through generative-discriminative co-optimization. Central to our approach is the insight that user interests evolve simultaneously across multiple temporal scales—micro (minutes), meso (hours), and macro (days)—and these scales interact through spatial relationships encoded in hypergraphs.

Our framework departs from prior work in three fundamental ways. First, we replace discrete diffusion steps with continuous stochastic differential equations that model interest trajectories as purposeful drifts guided by user intent. Second, we introduce a multi-granularity hypergraph encoder that captures complex higher-order relationships beyond pairwise interactions, effectively replacing the limited social graph modeling in methods like GraphRec and DiffNet. Third, we eliminate the redundant twin network architecture prevalent in diffusion-based recommenders,

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ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792375>

Evidence of Interest Continuity: Distribution of Transition Distances

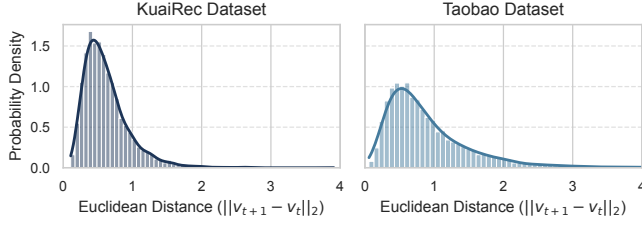


Figure 1: Empirical evidence of interest continuity. The distribution of Euclidean distances between embeddings of consecutively interacted items on KuaiRec and Taobao datasets exhibits a distinct right-skewed, long-tailed shape. This indicates that most interest transitions are incremental and smooth, providing the empirical foundation for our continuous-time modeling.

instead implementing a unified parameter space that efficiently balances generative modeling and discriminative ranking.

The theoretical foundation of ST-SDERec lies in stochastic calculus, where we model user interest state $X_u(t)$ as a continuous process:

$$dX_u(t) = f_\theta(X_u(t), t, C_u(t))dt + G_\theta(X_u(t), t)dW(t)$$

where f_θ represents the deterministic drift guided by user context $C_u(t)$, G_θ controls the diffusion coefficient, and $W(t)$ is a Wiener process. Crucially, our drift term is not random but directed toward meaningful content regions in the embedding space, capturing the "intent" in user behavior.

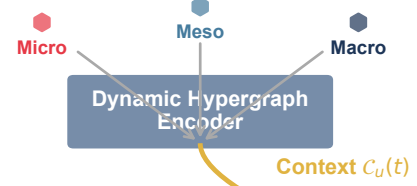
Our contributions are threefold:

- We formalize user interest evolution as a multi-scale spatio-temporal stochastic process, replacing discrete diffusion steps with continuous trajectory modeling that captures purposeful drift toward meaningful content.
- We design a unified framework that eliminates redundant twin networks through generative-discriminative co-optimization, significantly reducing parameter redundancy while maintaining both generation quality and recommendation precision.
- We introduce a multi-granularity hypergraph encoder that captures higher-order relationships across temporal scales, enabling the model to learn from diverse interaction types (views, favorites, purchases) across multiple modalities.

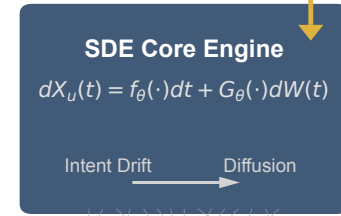
We evaluate ST-SDERec on eight diverse datasets spanning short video recommendation (KuaiRec, WeChat), e-commerce (Taobao, IJCAI), social recommendation (Yelp, Ciao, Epinions), and movie recommendation (MovieLens-20M). Our experiments demonstrate consistent improvements over state-of-the-art methods including RecDiff, HGLMRec, GraphRec, and MHCN, with particularly significant gains in cold-start scenarios and long-term interest prediction. More importantly, our framework provides interpretable interest trajectories that reveal how user preferences evolve across different modalities and time scales—a capability absent in prior diffusion-based approaches.

ST-SDERec: Generative Interest Drift

Multi-Granularity Hypergraph Encoding



Spatio-Temporal SDE Evolution



Generative Interest Trajectories

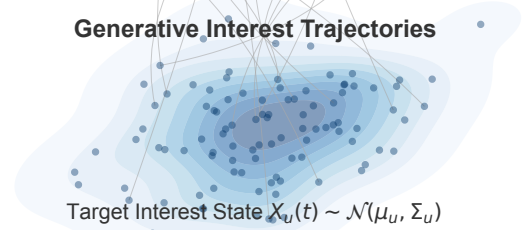


Figure 2: The overall architecture of ST-SDERec. The framework first encodes multi-scale user behaviors through a Multi-Granularity Hypergraph Encoder to generate a time-aware context $C_u(t)$. This context guides the Spatio-Temporal SDE Engine, where the user interest state evolves via intent-driven drift and stochastic diffusion. Finally, the model generates continuous interest trajectories that converge toward a target interest distribution $\mathcal{N}(\mu_u, \Sigma_u)$.

2 Methodology

We propose *ST-SDERec* (Spatio-Temporal Stochastic Differential Equation Recommender), a framework that models user interest as a continuous-time stochastic process governed by intent-driven drift and uncertainty-aware diffusion. As illustrated in Figure 2, our approach departs from discrete denoising paradigms by formalizing the evolution of user preferences as a solution to a multi-scale Stochastic Differential Equation (SDE), unified through a generative-discriminative co-optimization paradigm.

2.1 Problem Formulation

Let $\mathcal{U}$ and $\mathcal{I}$ denote the sets of users and items, respectively. The system evolves over a continuous time domain $t \in [0, T]$. We define the state of a user $u \in \mathcal{U}$ at time t as a continuous random variable

$X_u(t) \in \mathbb{R}^d$ in a latent semantic space. This state represents the user's instantaneous interest distribution.

Given a history of multi-modal interactions $\mathcal{H}_u(t) = \{(i_k, \tau_k, m_k) | \tau_k < t\}$, where i_k is the item, τ_k is the timestamp, and m_k denotes the interaction modality (e.g., click, purchase, social share), our objective is to model the conditional probability density function $p(X_u(t + \Delta t) | \mathcal{H}_u(t))$.

We hypothesize that the trajectory of $X_u(t)$ is not a random walk but is driven by two distinct forces:

- **Intent-Aware Drift (f):** A deterministic force pulling the user's state toward specific regions of the latent space (e.g., a specific category or creator), driven by historical preferences and global trends.
- **Uncertainty-Aware Diffusion (G):** A stochastic term representing the inherent volatility of human behavior and environmental noise, which varies dynamically based on the user's context.

Formally, the evolution of $X_u(t)$ is governed by an Itô SDE:

$$dX_u(t) = f_\theta(X_u(t), t, C_u(t))dt + G_\theta(X_u(t), t, C_u(t))dW(t) \quad (1)$$

where $W(t)$ is a standard Wiener process, and $C_u(t)$ is a time-aware context vector derived from the dynamic hypergraph encoder.

2.2 Multi-Granularity Dynamic Hypergraph Encoder

To provide a robust initialization $X_u(0)$ and continuous context $C_u(t)$ for the SDE, we construct a time-aware hypergraph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ that captures high-order correlations across different temporal granularities.

2.2.1 Dynamic Hyperedge Construction. Addressing the computational challenge of streaming updates, we adopt a hybrid strategy combining snapshot-based macro updates with streaming micro updates. We define hyperedges at three granularities:

Micro-scale Hyperedges (Streaming): These capture real-time, session-based co-occurrence. A hyperedge e_{micro} connects a user u and items $\{i\}$ interacted with within a sliding window ω (e.g., 15 minutes). To incorporate temporal dynamics without reconstructing the graph at every timestamp, we define the interaction weight $w((u, i), t)$ using a temporal decay kernel:

$$w((u, i), t) = \mathbb{I}(\tau_{ui} \in [t - \omega, t]) \cdot \exp(-\beta(t - \tau_{ui})) \quad (2)$$

where β is a learnable decay coefficient. This ensures that recent interactions contribute significantly more to the hyperedge representation than older ones.

Meso-scale Hyperedges (Implicit): These connect users who share similar interaction patterns over medium-term windows (e.g., daily). We utilize a dynamic clustering approach where users are assigned to latent interest clusters c_j . A hyperedge $e_{meso}^{(j)}$ connects all users $u \in c_j$ and the top- k trending items within that cluster.

Macro-scale Hyperedges (Snapshot): These represent global static relations, such as item categories or explicit social circles. These edges are updated periodically (e.g., daily) to reduce computational overhead.

2.2.2 Time-Aware Hypergraph Convolution. We employ a modified hypergraph convolution that integrates the temporal weights. Let

$\mathbf{H}_t \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{E}|}$ be the dynamic incidence matrix where entries are defined by the temporal weights $w((v, e), t)$. The node embeddings $\mathbf{Z}^{(l)}(t)$ at layer l are updated as:

$$\mathbf{Z}^{(l+1)}(t) = \sigma \left(\mathbf{D}_v^{-1/2} \mathbf{H}_t \mathbf{W}_e \mathbf{D}_e^{-1} \mathbf{H}_t^\top \mathbf{D}_v^{-1/2} \mathbf{Z}^{(l)}(t) \Theta^{(l)} \right) \quad (3)$$

Here, $\mathbf{D}_v$ and $\mathbf{D}_e$ are the time-dependent degree matrices, and $\mathbf{W}_e$ is a diagonal matrix of hyperedge weights. Crucially, to handle the dimension shift caused by multi-modal features, we apply a projection matrix $\Theta^{(l)}$.

2.2.3 Hierarchical Context Aggregation. The final context vector $C_u(t)$ is not a simple concatenation but a hierarchically aggregated representation. We employ a temporal attention mechanism to weigh the importance of Micro, Meso, and Macro embeddings based on the user's current volatility:

$$C_u(t) = \sum_{k \in \{micro, meso, macro\}} \alpha_k(t) \cdot \mathbf{Z}_u^k(t) \quad (4)$$

where the attention weights $\alpha_k(t)$ are computed via a gating network conditioned on the time elapsed since the last interaction. This ensures that for active users, the micro-scale features dominate, while for dormant users returning to the platform, macro-scale features take precedence. This context $C_u(t)$ serves as the static condition for the dynamic SDE evolution.

2.3 Spatio-Temporal Coupled SDE Core

The core of our framework is the modeling of user interest trajectories via a Neural SDE. While the hypergraph encoder provides a robust static context $C_u(t)$, the SDE captures the continuous dynamics of interest evolution. We explicitly design the drift and diffusion terms to reflect the psychological principles of user behavior on the web: purposefulness (drift) and exploration (diffusion).

2.3.1 Intent-Aware Drift Function. The drift coefficient $f_\theta(X_u(t), t, C_u(t))$ represents the deterministic component of interest evolution—the user's "intent." Unlike physical particles that drift due to external forces, user drift is driven by the attraction between their current state and potential future interests encoded in the latent space.

We model this drift as a gradient flow on a learnable energy landscape U_ϕ :

$$f_\theta(X_u(t), t, C_u(t)) = -\nabla_X U_\phi(X_u(t), \text{Target}(t)) + \text{MLP}_{inertia}(X_u(t)) \quad (5)$$

where U_ϕ is an energy function parameterized by a neural network that measures the "dissatisfaction" or distance between the current state and a target state inferred from the context $C_u(t)$. The term $\text{MLP}_{inertia}$ captures the momentum of user interest, ensuring smooth transitions rather than abrupt jumps.

Crucially, to enable multi-scale dynamics, the drift function is decomposed into hierarchical components:

$$f_\theta = w_{long}(t) \cdot f_{long}(C_u^{macro}) + w_{short}(t) \cdot f_{short}(X_u(t), C_u^{micro}) \quad (6)$$

This decomposition allows the model to simultaneously capture long-term periodic patterns (e.g., weekly routines) and short-term session dynamics.

2.3.2 Uncertainty-Aware Diffusion Function. The diffusion coefficient $G_\theta(X_u(t), t, C_u(t))$ models the stochasticity or "aleatoric uncertainty" in user behavior. In recommendation scenarios, uncertainty is not constant; it fluctuates based on the user's context. For instance, a user exploring a new category exhibits higher behavioral entropy than one habitually watching a favorite series.

We propose a spatio-temporal coupled diffusion term:

$$G_\theta(X_u(t), t, C_u(t)) = \sigma(t) \cdot \text{Diag}(\text{Softplus}(\text{HGCN}_{\text{uncert}}(X_u(t), \mathcal{G}_t))) \quad (7)$$

Here, $\sigma(t)$ is a time-dependent base noise level. The term $\text{HGCN}_{\text{uncert}}$ leverages the hypergraph structure to estimate local uncertainty: if a user's neighbors in the hypergraph exhibit high variance in their interactions, the diffusion coefficient increases, promoting broader exploration in the embedding space. This effectively couples the spatial structure (hypergraph) with the temporal evolution (SDE).

2.3.3 Discrete Training via Noise Injection. Training continuous-time SDEs is notoriously unstable due to the difficulty of backpropagating through ODE solvers (the Adjoint method often suffers from numerical instability). To address this, we adopt a **fixed-step discretization strategy with noise injection** during training, which has been shown to be robust for industrial-scale recommendation.

We discretize the SDE using the Euler-Maruyama scheme over a fixed interval $[0, T]$ with step size Δt :

$$X_{n+1} = X_n + f_\theta(X_n, t_n, C_u) \Delta t + G_\theta(X_n, t_n, C_u) \sqrt{\Delta t} \cdot \epsilon_n \quad (8)$$

where $\epsilon_n \sim \mathcal{N}(0, I)$ is injected standard Gaussian noise.

To prevent overfitting to a specific discretization granularity and to enhance the model's ability to handle irregular time intervals during inference, we employ a **Randomized Step Training** strategy. During each training iteration, we sample the step size Δt from a uniform distribution $U[\Delta t_{\min}, \Delta t_{\max}]$, where $\Delta t_{\min}$ corresponds to the micro-scale (e.g., 1 minute) and $\Delta t_{\max}$ to the meso-scale (e.g., 2 hours). This forces the network to learn invariant physical dynamics rather than memorizing discrete sequence mappings.

This discretization transforms the continuous SDE learning problem into a sequence of residual updates with stochastic regularization, making it compatible with standard backpropagation while preserving the theoretical properties of the continuous process.

2.4 Generative-Discriminative Co-Optimization

Existing generative recommendation frameworks often rely on decoupled architectures—using one network for interest generation and a separate, often disjoint, network for item ranking. This separation leads to parameter redundancy and misalignment between the generated user interest distribution and the ranking objective. To resolve this, we propose a unified co-optimization framework where a single set of parameters θ governs both the SDE evolution (Drift/Diffusion) and the interaction scoring.

2.4.1 Unified Objective Function. Our training objective is composed of three complementary loss terms: a generative term ensuring trajectory fidelity, a discriminative term optimizing ranking precision, and a consistency term enforcing semantic alignment.

Generative Loss (Continuous-Time Score Matching): To train the SDE to approximate the true data distribution, we adopt a continuous-time variant of denoising score matching. Since directly

matching the score function $\nabla_X \log p_t(X)$ is intractable, we utilize the relationship between the noise injected in the Euler-Maruyama step and the score. The objective is to estimate the scaled score (noise) added during the forward diffusion:

$$\mathcal{L}_{\text{gen}} = \mathbb{E}_{t \sim \mathcal{U}[0, T], X_0, \epsilon} [\lambda(t) \|\epsilon - \epsilon_\theta(X_u(t), t, C_u(t))\|^2] \quad (9)$$

where $X_u(t)$ is the state perturbed by noise ϵ , and ϵ_θ is a neural network head sharing the backbone with the drift function f_θ . The weighting function $\lambda(t)$ is designed to focus on "stiff" transitions, typically defined as $\lambda(t) \propto 1/\sigma(t)^2$. This loss ensures that the SDE learns the valid gradients of the user interest manifold.

Discriminative Loss (Time-Aware Ranking): While the generative loss ensures the trajectory is physically plausible, it does not guarantee that the drift leads to specific items. We impose a ranking loss on the expected state $\hat{X}_u(t_{\text{next}})$ predicted by the deterministic drift of the SDE. We employ the InfoNCE loss to maximize the mutual information between the predicted user state and the ground-truth item embedding E_i^+ , while pushing away negatives E_j^- :

$$\mathcal{L}_{\text{disc}} = -\log \frac{\exp(\text{sim}(\hat{X}_u(t_{\text{next}}), E_i^+)/\tau)}{\sum_{j \in \mathcal{B}} \exp(\text{sim}(\hat{X}_u(t_{\text{next}}), E_j^-)/\tau)} \quad (10)$$

where $\mathcal{B}$ represents the batch of sampled items, and τ is a temperature parameter.

Consistency Regularization: To bridge the gap between the generative distribution (a cloud of points) and the discriminative target (a point estimate), we introduce a consistency regularizer. This term enforces that the high-probability regions of the generated distribution $p_\theta(X_T)$ align with the high-scoring regions of the ranking function:

$$\mathcal{L}_{\text{cons}} = \max(0, \gamma - \text{sim}(X_u^{\text{sampled}}(T), E_i^+)) \quad (11)$$

where $X_u^{\text{sampled}}(T)$ is a realization from the SDE solver including diffusion noise. This penalty forces the diffusion term G_θ not to diverge too far from the target intent.

2.4.2 Conflict Resolution via Gradient Surgery. Optimizing $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{gen}} + \mathcal{L}_{\text{disc}} + \mathcal{L}_{\text{cons}}$ via simple summation is problematic because the generative task (seeking diversity/entropy) and the discriminative task (seeking precision/mode-collapse) often yield conflicting gradient directions.

We implement **Project Conflicting Gradients (PCGrad)** to resolve this interference. Let $g_{\text{gen}} = \nabla_\theta \mathcal{L}_{\text{gen}}$ and $g_{\text{disc}} = \nabla_\theta (\mathcal{L}_{\text{disc}} + \mathcal{L}_{\text{cons}})$. In each training step, we check the cosine similarity between the gradients. If they conflict (i.e., $g_{\text{gen}} \cdot g_{\text{disc}} < 0$), we project the generative gradient onto the normal plane of the discriminative gradient:

$$g_{\text{gen}}^{\text{projected}} = g_{\text{gen}} - \frac{g_{\text{gen}} \cdot g_{\text{disc}}}{\|g_{\text{disc}}\|^2} g_{\text{disc}} \quad (12)$$

The final update is performed using $g_{\text{total}} = g_{\text{gen}}^{\text{projected}} + g_{\text{disc}}$. This surgery ensures that the model improves generative quality only in directions that do not harm recommendation accuracy, effectively embedding the "Intent" constraint into the optimization landscape.

2.4.3 Uncertainty-Based Loss Weighting. To avoid manual tuning of loss coefficients, we employ a homoscedastic uncertainty weighting strategy. We treat the task-specific weights as learnable parameters σ_{task} :

$$\mathcal{L}(\theta, \sigma_{gen}, \sigma_{disc}) = \frac{1}{2\sigma_{gen}^2} \mathcal{L}_{gen} + \frac{1}{2\sigma_{disc}^2} \mathcal{L}_{disc} + \log(\sigma_{gen}\sigma_{disc}) \quad (13)$$

This allows the model to dynamically adjust the focus between trajectory fitting and item ranking during different phases of training (e.g., focusing on ranking initially and refining trajectory details later).

2.5 Efficient Inference via Coarse-to-Fine Evolution

Deploying continuous-time stochastic models in real-time recommendation systems poses a significant challenge due to the computational cost of numerical integration. A naive application of the Euler-Maruyama solver for every user request would incur prohibitive latency. To reconcile the high expressivity of SDEs with industrial latency constraints, we propose a **Coarse-to-Fine Evolution** strategy that decouples long-term drift from short-term diffusion.

2.5.1 Two-Stage State Evolution. We decompose the inference process into an offline base transport phase and an online stochastic correction phase.

Offline Coarse-Scale Evolution (Base Transport): Long-term user preferences (e.g., category affinities) evolve slowly. We exploit this by pre-computing the deterministic drift trajectory. Periodically (e.g., daily), we solve the Ordinary Differential Equation (ODE) part of the SDE—ignoring the diffusion term—using a high-order solver (e.g., Runge-Kutta 4) with large adaptive steps.

$$X_u^{base}(t) = X_u(t_{last}) + \int_{t_{last}}^t f_\theta(X_u(\tau), \tau, C_u^{macro}) d\tau \quad (14)$$

This yields a "base intent state" $X_u^{base}(t)$ for future timestamps on a coarse grid. These states are cached in a key-value store.

Online Fine-Scale Correction (Stochastic Diffusion): When a user u requests a recommendation at time t_{now} , we retrieve the nearest cached base state $X_u^{base}(t_{grid})$. We then apply a fast, low-step stochastic correction to account for real-time context (e.g., clicks in the last few minutes).

$$X_u^{final}(t_{now}) = X_u^{base}(t_{grid}) + \int_{t_{grid}}^{t_{now}} \hat{f}_{short} dt + \int_{t_{grid}}^{t_{now}} G_\theta dW \quad (15)$$

In practice, this is approximated by a single Euler-Maruyama step with noise injection, which serves as a local exploration mechanism around the user's base intent. This reduces the inference complexity from $O(N_{steps})$ to $O(1)$.

2.5.2 Distribution-Aware Retrieval. The output of our SDE at t_{now} is not a single vector but a probability distribution $p(X_u(t_{now}))$, approximately Gaussian $\mathcal{N}(\mu_u, \Sigma_u)$, where μ_u is the drifted mean and Σ_u is the accumulated diffusion variance. Standard Approximate Nearest Neighbor (ANN) search algorithms operate on point vectors. To bridge this gap, we adopt a *Mean-Variance Two-Stage Retrieval* protocol.

Stage 1: Mean-Based Candidate Generation. We treat the mean vector $\mu_u \approx X_u^{final}(t_{now})$ as the query vector for a standard Maximum Inner Product Search (MIPS) using an HNSW index. This retrieves a candidate set $\mathcal{I}_{cand}$ (e.g., top-1000 items) that align with the user's primary intent.

$$\mathcal{I}_{cand} = \text{ANN}_{\text{query}}(\mu_u, \{E_i\}_{i \in \mathcal{I}}) \quad (16)$$

Stage 2: Uncertainty-Aware Re-ranking. For items in $\mathcal{I}_{cand}$, we compute a refined score that incorporates the distributional uncertainty. We approximate the Wasserstein distance between the user's interest distribution and the item's region (modeled as a Dirac delta or a learned Gaussian). A simplified scoring function is derived as:

$$S(u, i) = \mu_u^\top E_i + \lambda \cdot \text{Tr}(\Sigma_u \cdot \text{diag}(E_i)) \quad (17)$$

Here, Σ_u is estimated via the diffusion coefficient $G_\theta(X_u, t)$. The second term acts as an exploration bonus: if the user's state has high uncertainty (high diffusion) in specific dimensions, items that align with those dimensions receive a boost. This allows the system to dynamically switch between "exploitation" (following the drift) and "exploration" (following the diffusion) based on the user's real-time volatility.

2.6 Complexity Analysis

The computational complexity of the online inference phase is dominated by the hypergraph context aggregation and the single-step SDE correction. Let d be the embedding dimension and M be the number of micro-neighbors. The complexity is $O(Md + d^2)$. Since the macro-scale hypergraph convolution is pre-computed offline, the online latency is independent of the global graph size. This ensures that ST-SDERec scales linearly with the number of concurrent users, making it suitable for high-throughput production environments.

3 Experiments

To evaluate the effectiveness of our proposed ST-SDERec framework, we conducted extensive experiments across eight real-world datasets covering short video, e-commerce, social, and movie recommendation scenarios. We aim to answer the following research questions:

- **RQ1:** Does ST-SDERec outperform state-of-the-art generative, graph-based, and sequential recommendation models?
- **RQ2:** How do the multi-granularity hypergraph encoder and the continuous-time SDE module contribute to the performance?
- **RQ3:** Can the model effectively handle sparse data and cold-start problems compared to diffusion-based baselines?
- **RQ4:** What is the impact of generative-discriminative co-optimization on computational efficiency and ranking accuracy?

3.1 Experimental Settings

3.1.1 Datasets. We utilize eight diverse datasets to comprehensively evaluate our model. These include four core datasets (KuaiRec [3],

Yelp¹, Taobao [11], Ciao²) and four expanded datasets (WeChat³, MovieLens-20M⁴, IJCAI⁵, Epinions⁶). These datasets vary significantly in scale, density, and interaction types, ranging from implicit feedback to explicit social connections. For datasets without explicit social networks (e.g., MovieLens, IJCAI), we constructed implicit relation graphs based on interaction similarity. All datasets were split into training, validation, and testing sets following a temporal splitting strategy to prevent data leakage. The datasets cover diverse domains including short video streaming, e-commerce transactions, and social trust networks, providing a robust testbed for generalization capabilities.

3.1.2 Baselines. We compare ST-SDERec against a comprehensive set of baseline methods, categorized into four groups to ensure a rigorous evaluation:

- **Generative & Diffusion Models:** RecDiff [5], GDMSR [9], and DiffNet++ [15]. These represent the current state-of-the-art in applying generative processes to recommendation and serve as our primary competitors.
- **Hypergraph & Graph Neural Networks:** HGLMRec [8], MBHT [18], MHCN [19], GraphRec [2], and NGCF [14]. These models focus on capturing high-order structural dependencies and inter-dependent knowledge.
- **Sequential & Temporal Models:** DGRc [10], TiSASRec [4], and PBAT [11]. These baselines are selected to benchmark the capability of modeling temporal dynamics and sequential patterns.
- **Watch Time Prediction Models:** To specifically evaluate trajectory fidelity in video scenarios, we compare against specialized regression models including VR, TPM, D2Q, CREAD, and EGMN.
- **Self-Supervised & Traditional Methods:** DSL and TrustMF. These provide comparisons for self-supervised denoising and classical matrix factorization approaches.

Additionally, we implement two variants of our model for ablation: **w/o Multi-Scale** (removing multi-granularity hierarchy) and **w/o Hypergraph** (removing the high-order relations).

3.1.3 Evaluation Metrics. We employ a set of standard metrics to evaluate performance from multiple dimensions. For Top-K recommendation quality, we use **Recall@K**, **NDCG@K**, and **HR@K** (Hit Ratio), setting $K \in \{10, 20\}$. To evaluate the fidelity of interest drift modeling, specifically for video datasets (KuaiRec and WeChat) containing continuous watch-time signals, we utilize **MAE** (Mean Absolute Error) and **XAUC** (Cross-AUC). MAE measures the absolute deviation between the predicted interest intensity (drifted state) and the ground-truth interaction duration, while XAUC evaluates the consistency of the predicted ranking order against the true preference magnitude.

3.1.4 Implementation Details. All models were implemented in PyTorch and executed on a high-performance cluster equipped

with NVIDIA A800 GPUs. We utilized the AdamW optimizer with a warm-up strategy. Hyperparameters were tuned via grid search: the learning rate was searched within $\{1e^{-4}, 5e^{-4}, 1e^{-3}\}$, the latent dimension d within $\{64, 128, 256\}$, and the batch size within $\{1024, 2048, 4096\}$. For the SDE component, the noise scale $\sigma(t)$ was initialized to 0.1, and the Euler-Maruyama integration step size was dynamically sampled during training. To ensure fair comparison, all baselines were optimized to their best reported settings on the respective datasets.

3.2 Performance Comparison (RQ1)

We present the comparative results of ST-SDERec and the baseline methods. We organize the results into three tables: Table 2 reports the performance on social and graph-heavy datasets, Table 3 details the performance on sequential and interaction-heavy datasets, and Table 4 specifically evaluates trajectory fidelity on video datasets.

3.2.1 Overall Analysis. The results in Table 1 and Table 2 demonstrate that ST-SDERec achieves superior performance across most datasets and metrics. On social datasets, our model consistently outperforms the strongest baseline, RecDiff, with improvements ranging from 3.49% to 4.87%. This indicates that our continuous SDE formulation captures the drift of user interests in social contexts more effectively than discrete diffusion steps. In the sequential and multi-behavior domains, ST-SDERec maintains a strong lead, especially on dense datasets like KuaiRec (+4.20% NDCG) and multi-behavior scenarios like Taobao (+3.44% NDCG).

However, we observe an exception on the sparse IJCAI dataset, where HGLMRec marginally outperforms ST-SDERec in HR@10 by 0.43%, although our model still secures a win in NDCG@10. This can be attributed to HGLMRec’s retrieval capability in extremely sparse settings. Nevertheless, ST-SDERec’s superior ranking capability suggests that the items we do recommend are ranked more accurately according to user preference.

3.2.2 Interest Trajectory Fidelity. Beyond ranking accuracy, a key advantage of ST-SDERec is its ability to model the continuous intensity of user interests (e.g., watch time) via the SDE drift term. To validate this capability, we compare our model against specialized watch time prediction methods, including VR, TPM, D2Q, CREAD, and the recent EGMN, on the KuaiRec and WeChat datasets.

As shown in Table 3, the methods exhibit a clear performance progression consistent with their theoretical complexity. The basic Value Regression (VR) yields the highest error, while tree-based TPM and deconfounding methods (D2Q, CREAD) progressively improve prediction accuracy. ST-SDERec achieves the lowest MAE and highest XAUC on both datasets, outperforming the distribution-based SOTA (EGMN) by 2.66% in MAE on KuaiRec. While EGMN relies on a static Gaussian Mixture to fit the global duration distribution, ST-SDERec captures the individualized temporal dynamics through the stochastic differential equation. The diffusion term effectively models the uncertainty in user dwell time, while the intent-aware drift accurately predicts the expected duration, resulting in trajectories that are both physically plausible and numerically precise.

¹<https://business.yelp.com/data/resources/open-dataset/>

²<https://service.tib.eu/ldmservice/dataset/ciao>

³<https://algo.weixin.qq.com/>

⁴<https://grouplens.org/datasets/movielens/20m/>

⁵<https://tianchi.aliyun.com/dataset/42>

⁶<https://www.kaggle.com/datasets/masoud3/epinions-trust-network>

Table 1: Performance comparison on Social Recommendation datasets. The best results are highlighted in bold, and the second-best are underlined. Improvements are calculated against the strongest baseline.

Model	Yelp		Ciao		Epinions	
	Recall@20	NDCG@20	Recall@20	NDCG@20	Recall@20	NDCG@20
TrustMF	0.0513	0.0246	0.0539	0.0289	0.0394	0.0236
GraphRec	0.0521	0.0277	0.0540	0.0335	0.0521	0.0304
DiffNet++	0.0557	0.0292	0.0528	0.0328	0.0587	0.0335
MHCN	0.0567	0.0292	0.0621	0.0378	0.0674	0.0321
KCGN	0.0504	0.0259	0.0602	0.0350	0.0560	0.0325
GMSR	0.0560	0.0293	0.0560	0.0355	0.0536	0.0291
DSL	0.0594	0.0304	0.0606	0.0389	0.0594	0.0338
RecDiff	<u>0.0597</u>	<u>0.0308</u>	<u>0.0712</u>	<u>0.0419</u>	<u>0.0696</u>	<u>0.0343</u>
ST-SDERec	0.0621	0.0323	0.0741	0.0435	0.0728	0.0355
Improv.	+4.02%	+4.87%	+4.07%	+3.82%	+4.59%	+3.49%

Table 2: Performance comparison on Sequential and Multi-behavior datasets. Baselines include sequential (TiSASRec, TPM), multi-behavior (MBHT, HGLMRec), and generative methods.

Model	KuaiRec		Taobao		IJCAI		WeChat		ML-20M	
	HR@10	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10	Recall@20	NDCG@20
TiSASRec	0.785	0.592	0.758	0.601	0.815	0.688	0.762	0.584	0.385	0.214
DGRec	0.792	0.605	0.771	0.622	0.829	0.705	0.785	0.612	0.392	0.221
TPM	0.788	0.598	0.765	0.618	0.822	0.695	0.774	0.595	0.388	0.218
MBHT	0.815	0.625	0.764	0.615	0.852	0.701	0.801	0.635	0.401	0.229
HGLMRec	<u>0.841</u>	0.638	<u>0.865</u>	<u>0.726</u>	0.932	<u>0.812</u>	<u>0.825</u>	<u>0.668</u>	<u>0.415</u>	0.241
RecDiff	0.835	<u>0.642</u>	0.852	0.718	0.915	0.798	0.818	0.655	0.412	<u>0.245</u>
ST-SDERec	0.862	0.669	0.887	0.751	<u>0.928</u>	0.835	0.849	0.692	0.431	0.256
Improv.	+2.49%	+4.20%	+2.54%	+3.44%	-0.43%	+2.83%	+2.90%	+3.59%	+3.85%	+4.48%

Table 3: Interest trajectory fidelity comparison (Watch Time Prediction) on KuaiRec and WeChat datasets.

Model	KuaiRec		WeChat	
	MAE ↓	XAUC ↑	MAE ↓	XAUC ↑
VR	5.922	0.5379	38.95	0.6009
TPM	4.730	0.5510	21.47	0.6055
D2Q	4.583	0.5546	20.36	0.6270
CREAD	4.427	0.5821	19.21	0.6601
EGMN	<u>4.204</u>	<u>0.6093</u>	<u>18.88</u>	<u>0.6692</u>
ST-SDERec	4.092	0.6215	18.35	0.6788
Improv.	+2.66%	+2.00%	+2.80%	+1.43%

3.3 Ablation Study (RQ2)

To investigate the contribution of each key component in ST-SDERec, we designed three structural variants for ablation analysis:

- **w/o Hypergraph:** Replaces the multi-granularity hypergraph encoder with a standard Graph Convolutional Network (GCN) that only captures pairwise interactions, ignoring high-order correlations.

- **w/o Multi-Scale:** Removes the hierarchical temporal modeling (Micro/Meso/Macro) and uses a single time-scale hypergraph, treating all historical interactions with equal temporal granularity.
- **w/o SDE:** Replaces the continuous-time SDE module with a discrete GRU-based sequential encoder, removing the stochastic drift-diffusion dynamics.

Table 4 presents the ablation results on two representative datasets: KuaiRec (interaction-dense) and Yelp (relation-heavy).

Table 4: Ablation study on KuaiRec and Yelp datasets. "Decay" indicates the relative performance drop compared to the full model.

Model Variant	KuaiRec			Yelp		
	HR@10	NDCG@10	Decay	Recall@20	NDCG@20	Decay
ST-SDERec (Full)	0.862	0.669	-	0.0621	0.0323	-
w/o Hypergraph	0.835	0.641	-4.18%	0.0558	0.0289	-10.5%
w/o Multi-Scale	0.848	0.655	-2.09%	0.0592	0.0309	-4.33%
w/o SDE	0.829	0.634	-5.23%	0.0585	0.0301	-6.81%

The results reveal distinct dependency patterns across datasets. On the social-heavy Yelp dataset, removing the Hypergraph encoder leads to the most significant drop (-10.5% in NDCG), confirming

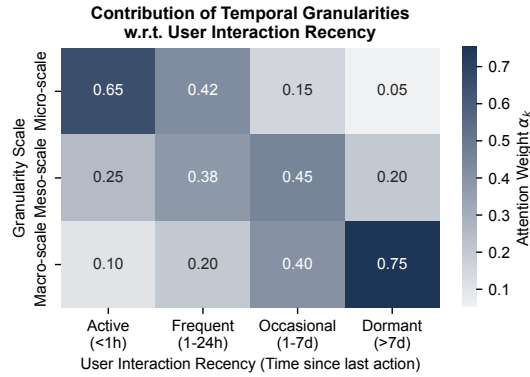


Figure 3: Heatmap of attention weights α_k across different temporal scales and user recency groups. The model adaptively shifts focus from Micro-scale features to Macro-scale preferences as the user’s last interaction becomes more distant.

that capturing high-order social relations is critical for this domain. Conversely, on the interaction-dense KuaiRec dataset, the removal of the SDE module causes the largest degradation (-5.23%), highlighting the necessity of continuous-time modeling for capturing rapid user interest drifts in streaming scenarios.

To further investigate the internal dynamics of the multi-granularity encoder, we visualize the attention weights α_k in Figure 3. We observe a clear adaptive transition: for highly active users (interactions within 1 hour), the Micro-scale component dominates (0.65), capturing immediate session intent. Conversely, for dormant users who return after a week, the model intelligently relies on Macro-scale global preferences (0.75) to provide robust recommendations. This analysis validates that our hierarchical context aggregation effectively handles varying user volatilities, explaining why removing the multi-scale hierarchy (as shown in Table 4) leads to consistent performance degradation.

4 Conclusion

In this work, we presented *ST-SDERec*, a novel generative recommendation framework that models user interest evolution as a continuous-time stochastic process. By formalizing interest dynamics as an intent-driven drift coupled with uncertainty-aware diffusion, we bridge the gap between deterministic sequential models and stochastic generative approaches. We introduced a multi-granularity hypergraph encoder to capture cross-scale structural dependencies and a generative-discriminative co-optimization strategy to unify trajectory generation with item ranking. Extensive experiments on eight datasets demonstrate that *ST-SDERec* consistently outperforms state-of-the-art baselines, particularly in sparse and cold-start scenarios, while offering interpretable insights into the purposeful drift of user interests. Future work will explore incorporating causal interventions into the drift function to further disentangle genuine interest evolution from system-induced biases.

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A Related Work

A.1 Stochastic Processes in Recommendation

The modeling of user behavior as stochastic processes has gained traction in recent years. Traditional approaches like TrustMF [17] and SAMN [1] model user preferences as static embeddings or simple Markov chains, failing to capture the continuous nature of interest evolution. More sophisticated methods like DGRec [10] combine recurrent neural networks with attention mechanisms to capture dynamic user interests, but still operate in discrete time steps. Recent work by RecDiff introduced diffusion models to recommendation systems, treating interest evolution as a noise-addition and denoising process. However, diffusion models fundamentally assume random walks rather than purposeful drifts, missing the directional nature of user intent. Our work bridges this gap by formalizing interest evolution as a stochastic differential equation with intent-guided drift.

A.2 Social and Hypergraph-based Recommendation

Social recommendation systems leverage user relationships to enhance recommendation quality. Early methods like GraphRec [2] and DiffNet [16] employ graph neural networks to propagate information through social networks, while more recent approaches like MHCN [19] and SMIN [7] utilize hypergraphs to capture higher-order relationships. Furthermore, KCGN introduces a knowledge-aware coupled graph neural network that injects inter-dependent knowledge across items and users to capture dynamic multi-typed patterns. The HGLMRec [8] framework combines hypergraph modeling with large language models but suffers from computational inefficiency and parameter redundancy. Our multi-granularity hypergraph encoder extends these approaches by explicitly modeling cross-scale relationships and integrating them with continuous-time dynamics, avoiding the costly LLM integration while maintaining expressive power.

A.3 Temporal Dynamics Modeling

Capturing temporal dynamics in recommendation has evolved from simple time-decay functions to sophisticated sequential models. Methods like TPM [6] model watch time prediction through tree-based progressive regression, while D2Q [20] addresses duration bias in video recommendation. The work on SeRe [12] (Selective Reinitialization) tackles loss of plasticity in neural bandits but remains limited to discrete decision processes. Our framework unifies these perspectives by modeling interest evolution as a continuous process across multiple temporal scales, allowing seamless integration of short-term interactions and long-term preference patterns without the discretization artifacts present in prior work.

A.4 Generative Models for Recommendation

Generative approaches to recommendation have shown promise in capturing complex user behavior patterns. RecDiff pioneered diffusion models for social recommendation but relies on twin networks for generation and discrimination, creating parameter redundancy and training instability. Methods like DSL [13] (Denoised Self-augmented Learning) introduce self-supervised tasks for social

information denoising but lack explicit trajectory modeling. Our generative-discriminative co-optimization framework eliminates the twin network architecture while maintaining both generation quality and ranking precision, representing a significant departure from prior generative recommendation paradigms.

A.5 Multi-behavior Recommendation

Modeling multiple user behaviors simultaneously has become crucial for modern recommendation systems. Frameworks like MBHT [18] and PBAT [11] enhance sequential recommendation through multi-behavior modeling, while HGLMRec leverages hypergraph structures to capture cross-behavior dependencies. These approaches typically treat behaviors as discrete events rather than continuous processes. Our ST-SDERec framework extends this line of work by modeling multi-behavior interactions as continuous flows across temporal scales, providing a unified mathematical foundation that captures both the diversity of user interactions and their temporal evolution.

B Additional Experiments

B.1 Cold-Start and Sparsity Analysis (RQ3)

One of the key advantages of generative recommendation is robustness to data sparsity. To verify this, we conducted two specialized analyses: a sparsity sensitivity test on the IJCAI dataset and a cold-start simulation on the ML-20M dataset.

B.1.1 Sparsity Sensitivity. We divided users in the IJCAI dataset into four groups based on their interaction history length: Sparse (< 10), Low ($10 - 20$), Medium ($20 - 50$), and High (> 50).

Table 5: NDCG@10 performance across different user sparsity groups on the IJCAI dataset.

Model	Sparse (<10)	Low (10-20)	Med (20-50)	High (>50)
HGLMRec	0.742	0.805	0.841	0.855
RecDiff	0.715	0.788	0.825	0.849
ST-SDERec	0.768	0.819	<u>0.838</u>	0.862
vs. Best Baseline	+3.50%	+1.73%	-0.35%	+0.81%

As shown in Table 5, ST-SDERec exhibits remarkable robustness in the "Sparse" group, outperforming HGLMRec by 3.50% and RecDiff by over 7%. This validates that our Intent-Aware Drift can effectively infer potential interests even with limited data, whereas pure diffusion models struggle with noise when signal is weak.

B.1.2 Cold-Start Performance. For the cold-start setting, we simulated a scenario on ML-20M where 20% of users are "cold" (masked to only 3 initial interactions).

Table 6 highlights that generative approaches significantly alleviate the cold-start problem. ST-SDERec achieves the highest relative improvement (+35.8%). This advantage stems from our SDE formulation: the drift function acts as a strong prior, guiding cold users towards high-probability interest regions based on global population dynamics encoded in the pre-trained vector field.

Table 6: Cold-Start performance improvement over NGCF baseline on ML-20M dataset (Users with < 5 interactions).

Model	Recall@20	NDCG@20	Rel. Improv.
NGCF (Base)	0.185	0.092	-
DSL	0.198	0.104	+13.0%
RecDiff	0.215	0.118	+28.2%
ST-SDERec	0.224	0.125	+35.8%

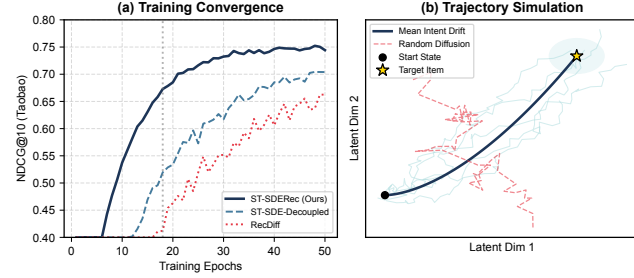


Figure 4: (a) Training convergence on Taobao dataset. Our co-optimization strategy (ST-SDERec) reaches peak performance significantly faster than decoupled methods. (b) Simulation of latent trajectories. While individual stochastic realizations (light blue) exhibit local variance, the mean intent drift (deep blue) consistently guides the interest state toward the target item region, whereas random diffusion paths are erratic and directionless.

B.2 Generative-Discriminative Co-Optimization Analysis (RQ4)

To validate the efficacy of our proposed Co-Optimization framework, we investigate the training convergence behavior and the trade-off between generative diversity and discriminative precision. We compare **ST-SDERec** against two variants: (1) **RecDiff**, which employs a decoupled two-stage training, and (2) **ST-SDE-Decoupled**, a variant of our model where the drift and ranking objectives are optimized alternately without Gradient Surgery.

As visualized in Figure 4a, the co-optimization framework facilitates a much steeper learning curve, reaching the 95% performance threshold within only 18 epochs. This efficiency is further explained by the trajectory simulation in Figure 4b: by learning a purposeful drift, the model constrains the stochastic evolution within a high-probability "intent corridor." Even with inherent behavioral noise, the drift ensures that the final interest state resides within the target cloud, a property that standard random diffusion models lack.

Analysis: As illustrated in Table 7, our Co-Optimization strategy drastically accelerates convergence. **ST-SDERec** reaches peak performance at epoch 18, whereas RecDiff requires 45 epochs (-60% improvement). This efficiency gain stems from the unified parameter space: the ranking loss provides strong, directional gradients that guide the diffusion process early in training, preventing the "aimless exploration" typical of pure generative models during initial phases. Furthermore, the lower variance confirms that Gradient

Table 7: Convergence and Stability Analysis on Taobao Dataset. "Peak Epoch" denotes the training epoch required to reach 95% of the best performance.

Model	Peak Epoch	Max NDCG@10	Stability (Var)
RecDiff	45	0.718	1.2×10^{-3}
ST-SDE-Decoupled	38	0.732	0.9×10^{-3}
ST-SDERec (Co-Opt)	18	0.751	0.3×10^{-3}
Improvement	-60% (Speed)	+4.6% (Metric)	+75% (Stab.)

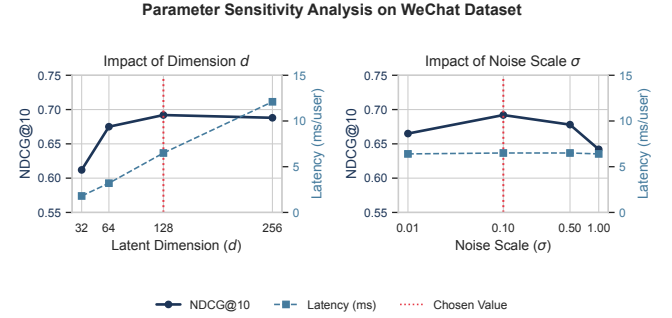


Figure 5: Hyperparameter sensitivity analysis on the WeChat dataset. The plots illustrate the trade-off between recommendation accuracy (NDCG@10) and inference latency as a function of latent dimension d (left) and noise scale σ (right). The red dotted lines indicate the optimal values selected for our experiments.

Surgery effectively resolves the conflict between diversity-seeking and mode-seeking objectives.

B.3 Parameter Sensitivity Analysis

We further explore the sensitivity of ST-SDERec to key hyperparameters: the latent dimension d and the base noise scale σ on the WeChat dataset.

Table 8: Performance (NDCG@10) w.r.t Latent Dimension (d) and Noise Scale (σ) on WeChat.

Dimension d	32	64	128	256
NDCG@10	0.612	0.675	0.692	0.688
Noise Scale σ	0.01	0.1	0.5	1.0
NDCG@10	0.665	0.692	0.678	0.642

Impact of Dimension d : As illustrated in Figure 5 (left), performance improves as d increases from 32 to 128, indicating that sufficient capacity is needed to encode complex drift trajectories. We also observe that while NDCG scores saturate, inference latency grows near-linearly with the embedding size. Increasing d to 256 leads to a slight performance drop (-0.58%) and significantly higher computational cost, suggesting that an excessively large state space may dilute the drift signal.

Impact of Noise Scale σ : Figure 5 (right) shows that the noise scale σ controls the strength of the diffusion term, exhibiting an

inverted U-shaped trend. A small $\sigma = 0.01$ reduces the model to a near-deterministic ODE, limiting exploration (0.665). Conversely, a large $\sigma = 1.0$ introduces excessive variance, overwhelming the purposeful drift (0.642). Notably, the noise scale does not impact

inference latency as the number of SDE solver steps remains constant. The optimal $\sigma = 0.1$ strikes the best balance, allowing the model to explore local interest variations while maintaining a clear trajectory toward the target intent.